

2025 NOV PURE MATHEMATICS PAPER 1 SOLUTIONS
6042/1 ADVANCED LEVEL

PURE MATHEMATICS
PAPER 1
6042/1

3 hours

ONLINE REVISION

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INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the spaces provided.
 - Answer all questions.
 - Numerical answers should be given to appropriate accuracy.
-

INFORMATION FOR CANDIDATES

- The number of marks is given in brackets [] at the end of each question.
-

QUESTIONS

1. The mass, m , of a soccer ball made of plastic of uniform density varies directly as the cube of its diameter, d . A soccer ball with diameter 7 cm has a mass of 0.11 kg. Calculate the mass of a similar soccer ball of diameter 9 cm. [4]

2. Given the differential equation

$$\frac{dy}{d\theta} + 2 \sin 2\theta = 0, \quad \text{where } y = 1, \quad \text{when } \theta = \frac{\pi}{6},$$

find the equation connecting y and θ . [4]

3. Solve the equation $z^2 + 3z + 4i + 2 = 0$, giving the answer in the form $x + iy$, where x and y are real. [5]

4. (a) Find, in terms of θ , an appropriate value of

$$\frac{\sin 4\theta + \tan 2\theta}{3 + \cos 2\theta},$$

for small values of θ , rad ($\theta \rightarrow 0$), neglecting terms in θ^2 and higher powers. [2]

- (b) Given that $1^\circ \approx 0.018$ radians and $(0.018)^2 \approx 0.000324$, find an approximate value for

$$\frac{\cos 2^\circ}{\sin 1^\circ}.$$

Give the answer correct to 2 decimal places. [3]

5. Given that the equation

$$x^2 + 2x + 4 = P(x^2 + 4)$$

has equal roots, find the possible values of P . [5]

6. Prove that the transformation matrix $\begin{pmatrix} 4 & -1 \\ -3 & 2 \end{pmatrix}$ maps all points on the line $y = 3x$ onto themselves. [5]

7. Solve the equation

$$2(3^{2x+1}) - 20(3^x) = -6.$$

[6]

8. Prove by mathematical induction that

$$\frac{1}{2} + \frac{1}{6} + \cdots + \frac{1}{n(n-1)} = 1 - \frac{1}{n} \quad \text{for } n > 1.$$

[6]

9. The line LM satisfies the equation $2x - 14y + 6 = 0$. Points L and M have coordinates $(-3, 0)$ and $(k, 1)$ respectively.

Find the

- (a) value of k . [2]
- (b) length of LM, leaving your answer in surd form. [2]
- (c) equation of the line through M, which is parallel to the line $2x + y + 5 = 0$. [3]

10. Given that the equation of a curve is

$$y = 10x - x^2 - 27,$$

- (a) express $10x - x^2 - 27$ in the form $-(a - x)^2 + b$, where a and b are integers. [2]
- (b) Hence or otherwise show that $10x - x^2 - 27$ is always negative. [1]
- (c) State the coordinates of the maximum point of the curve $y = 10x - x^2 - 27$. [2]
- (d) Sketch the curve, showing the point where it crosses the y -axis. [3]

11. Given $f(x) = \frac{2x+5}{x+1}$, $x \in \mathbb{R}, x \neq -1$,

- (a) show that $f(x) \equiv 2 + \frac{3}{x+1}$. [2]
- (b) describe the sequence of transformations that map the graph of $y = \frac{1}{x}$ onto the graph of $y = f(x)$. [3]
- (c) sketch the graph of $y = f(x)$ for $x \geq -1$. [3]

12. By sketching suitable graphs, show that the equation

$$\cot x = 2x^2 - 1$$

has only one root in the interval $0 < x < \frac{\pi}{2}$. [3]

13. Given that $(G, *) = \{0, 1, 2, 3\}$.

Show whether $(G, *)$ is a group or not under addition modulo 4. [9]

14. Functions f and g are defined by

$$f : x \mapsto e^{3x}, \quad x \in \mathbb{R}$$

$$g : x \mapsto \ln(x+2), \quad x \in \mathbb{R}, \quad x > -2.$$

- (a) Express in terms of x ,
 - i. $f^{-1}(x)$, stating the domain.

ii. $fg(x)$. [6]

(b) Sketch on a single diagram, the graphs of $y = f(x)$ and $y = f^{-1}(x)$. [3]

15. (a) By sketching suitable graphs, on the same axes, show that the equation

$$e^{2x} + 4x - 5 = 0,$$

has exactly one root. [3]

(b) Show by calculation that the root of the equation in (a) lies between 0 and 1. [3]

(c) Use the Newton-Raphson method, starting with the value $x_0 = 0.5$, to find the root of

$$e^{2x} + 4x - 5 = 0,$$

correct to 2 decimal places. [5]

16. (a) Find the stationary points on the curve

$$y = 2x^3 + 3x^2 - 12x + 1$$

and determine their nature. [9]

(b) Hence sketch the curve in (a) showing clearly the position of the stationary points. [3]

17. It is given that $P = \begin{pmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & -2 & 3 \end{pmatrix}$.

(a) Find the inverse of P . [7]

(b) Hence or otherwise solve the following equations

$$x + y + 2z = 0$$

$$2x - y + z = -3$$

$$x - 2y + 3z = 5$$

[6]

SOLUTIONS

Question 1

Notes

This is a direct proportion problem where mass varies with the cube of diameter. Use the given data point to find the constant of proportionality, then apply it to find the new mass.

Important Formula: If $y \propto x^n$, then $y = kx^n$ where k is the constant of proportionality.

$$\text{Given: } m \propto d^3$$

$$m = kd^3 \quad (\text{where } k \text{ is constant})$$

$$\text{When } d = 7 \text{ cm}, m = 0.11 \text{ kg} :$$

$$0.11 = k(7)^3$$

$$k = \frac{0.11}{343} = \frac{11}{34300}$$

$$\text{For } d = 9 \text{ cm} :$$

$$m = \frac{11}{34300} \times 9^3$$

$$= \frac{11 \times 729}{34300}$$

$$= \frac{8019}{34300} \text{ kg}$$

$$\approx 0.23379... \text{ kg}$$

$$\approx \mathbf{0.23 \text{ kg}} \quad (2 \text{ significant figures})$$

Question 2

Notes

This is a first-order differential equation that can be solved by separating variables and integrating. Use the initial condition to find the constant of integration.

$$\frac{dy}{d\theta} + 2 \sin 2\theta = 0$$

$$\frac{dy}{d\theta} = -2 \sin 2\theta$$

$$\int dy = \int -2 \sin 2\theta d\theta$$

$$y = -2 \times \left(-\frac{1}{2}\right) \cos 2\theta + c$$

$$y = \cos 2\theta + c$$

Important Formula: $\int \sin(ax) dx = -\frac{1}{a} \cos(ax) + C$

Using initial condition: $y = 1$ when $\theta = \frac{\pi}{6}$

$$1 = \cos \left(2 \times \frac{\pi}{6}\right) + c$$

$$1 = \cos \frac{\pi}{3} + c$$

$$1 = \frac{1}{2} + c$$

$$c = \frac{1}{2}$$

$$y = \cos 2\theta + \frac{1}{2}$$

Question 3

Notes

Use the quadratic formula with complex coefficients. The discriminant will be complex, so we need to find its square root in the form $x + yi$.

$$z^2 + 3z + (4i + 2) = 0$$

$$\begin{aligned} \text{Using quadratic formula: } z &= \frac{-3 \pm \sqrt{3^2 - 4(1)(4i + 2)}}{2} \\ &= \frac{-3 \pm \sqrt{9 - 16i - 8}}{2} \\ &= \frac{-3 \pm \sqrt{1 - 16i}}{2} \end{aligned}$$

Finding $\sqrt{1-16i}$: Let $\sqrt{1-16i} = x + yi$, then:

$$(x + yi)^2 = 1 - 16i$$
$$x^2 - y^2 + 2xyi = 1 - 16i$$

Equating real and imaginary parts:

$$x^2 - y^2 = 1 \quad (1)$$
$$2xy = -16 \Rightarrow y = -\frac{8}{x} \quad (2)$$

Substitute (2) into (1):

$$x^2 - \left(-\frac{8}{x}\right)^2 = 1$$
$$x^2 - \frac{64}{x^2} = 1$$
$$x^4 - x^2 - 64 = 0$$
$$x^2 = \frac{1 \pm \sqrt{257}}{2}$$

Take positive: $x = \pm \sqrt{\frac{1 + \sqrt{257}}{2}}$

From (2): $y = -\frac{8}{x}$

Thus:

$$z = \frac{-3 \pm \left(\sqrt{\frac{1 + \sqrt{257}}{2}} - \frac{8}{\sqrt{\frac{1 + \sqrt{257}}{2}}} i \right)}{2}$$

$$z \approx -0.041 - 1.4i \quad \text{or} \quad -3.0 + 1.4i \quad (2 \text{ s.f.})$$

Question 4

Notes

Use small angle approximations: $\sin \theta \approx \theta$, $\tan \theta \approx \theta$, $\cos \theta \approx 1 - \frac{\theta^2}{2}$. Neglect terms of order θ^2 and higher.

Important Formula: Small angle approximations:

- $\sin \theta \approx \theta$
- $\tan \theta \approx \theta$
- $\cos \theta \approx 1 - \frac{\theta^2}{2}$

(a)

$$\begin{aligned}\frac{\sin 4\theta + \tan 2\theta}{3 + \cos 2\theta} &\approx \frac{4\theta + 2\theta}{3 + \left(1 - \frac{(2\theta)^2}{2}\right)} \\ &= \frac{6\theta}{4 - 2\theta^2} \\ &= \frac{6\theta}{4\left(1 - \frac{\theta^2}{2}\right)} \\ &\approx \frac{3\theta}{2} \quad (\text{neglecting } \theta^2 \text{ and higher terms})\end{aligned}$$

$$\frac{3\theta}{2}$$

(b) Using $\theta = 1^\circ \approx 0.018$ rad:

$$\begin{aligned}\frac{\cos 2^\circ}{\sin 1^\circ} &\approx \frac{1 - \frac{(2\theta)^2}{2}}{\theta} \\ &= \frac{1 - 2\theta^2}{\theta} \\ &= \frac{1 - 2(0.018)^2}{0.018} \\ &= \frac{1 - 0.000648}{0.018} \\ &= \frac{0.999352}{0.018} \\ &\approx 55.52\end{aligned}$$

55.52 (2 decimal places)

Question 5

Notes

Rearrange the equation to standard quadratic form, then use the condition $b^2 - 4ac = 0$ for equal roots.

$$\begin{aligned}x^2 + 2x + 4 &= P(x^2 + 4) \\x^2 + 2x + 4 &= Px^2 + 4P \\x^2 - Px^2 + 2x + 4 - 4P &= 0 \\(1 - P)x^2 + 2x + (4 - 4P) &= 0\end{aligned}$$

Important Formula: For $ax^2 + bx + c = 0$ to have equal roots: $b^2 - 4ac = 0$

For equal roots:

$$\begin{aligned}(2)^2 - 4(1 - P)(4 - 4P) &= 0 \\4 - 4(1 - P)(4 - 4P) &= 0 \\4 - (16 - 32P + 16P^2) &= 0 \\4 - 16 + 32P - 16P^2 &= 0 \\-16P^2 + 32P - 12 &= 0 \\4P^2 - 8P + 3 &= 0 \\(2P - 3)(2P - 1) &= 0\end{aligned}$$

$$P = \frac{3}{2} \quad \text{or} \quad P = \frac{1}{2}$$

Question 6

Notes

A line is invariant if all points on it map to points on the same line. Check if for points on $y = 3x$, the transformed coordinates also satisfy $y = 3x$.

Let $M = \begin{pmatrix} 4 & -1 \\ -3 & 2 \end{pmatrix}$. For any point (x, y) :

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = M \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4x - y \\ -3x + 2y \end{pmatrix}$$

For points on $y = 3x$, substitute $y = 3x$:

$$x' = 4x - (3x) = x$$

$$y' = -3x + 2(3x) = -3x + 6x = 3x$$

Thus $x' = x$ and $y' = 3x = y$, so the transformed point (x', y') lies on the same line $y = 3x$.

Question 7

Notes

Recognize that $3^{2x+1} = 3 \cdot 3^{2x} = 3 \cdot (3^x)^2$. Substitute $y = 3^x$ to convert to a quadratic equation.

$$2(3^{2x+1}) - 20(3^x) = -6$$

$$2(3 \cdot 3^{2x}) - 20(3^x) + 6 = 0$$

$$6(3^{2x}) - 20(3^x) + 6 = 0$$

Let $y = 3^x$, then $3^{2x} = y^2$:

$$6y^2 - 20y + 6 = 0$$

$$3y^2 - 10y + 3 = 0$$

$$(3y - 1)(y - 3) = 0$$

Thus:

$$y = \frac{1}{3} \quad \text{or} \quad y = 3$$

$$3^x = \frac{1}{3} \quad \text{or} \quad 3^x = 3$$

$$3^x = 3^{-1} \quad \text{or} \quad 3^x = 3^1$$

$$x = -1 \quad \text{or} \quad x = 1$$

$$\mathbf{x = -1 \quad \text{or} \quad x = 1}$$

Question 8

Notes

Prove base case ($n = 2$), assume true for $n = k$, then prove for $n = k + 1$ using the assumption.

Let $P(n)$ be the statement:

$$\frac{1}{2} + \frac{1}{6} + \cdots + \frac{1}{n(n-1)} = 1 - \frac{1}{n} \quad \text{for } n > 1$$

Base Case ($n = 2$):

$$\text{LHS} = \frac{1}{2(2-1)} = \frac{1}{2}$$

$$\text{RHS} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$\text{LHS} = \text{RHS}$$

Inductive Hypothesis: Assume $P(k)$ is true for some $k \geq 2$:

$$\frac{1}{2} + \frac{1}{6} + \cdots + \frac{1}{k(k-1)} = 1 - \frac{1}{k}$$

Inductive Step: Prove $P(k+1)$ is true:

$$\begin{aligned} \text{LHS for } P(k+1) &= \left[\frac{1}{2} + \frac{1}{6} + \cdots + \frac{1}{k(k-1)} \right] + \frac{1}{k(k+1)} \\ &= \left(1 - \frac{1}{k} \right) + \frac{1}{k(k+1)} \quad (\text{by hypothesis}) \\ &= 1 - \frac{k+1}{k(k+1)} + \frac{1}{k(k+1)} \\ &= 1 - \frac{k}{k(k+1)} \\ &= 1 - \frac{1}{k+1} \\ &= \text{RHS for } P(k+1) \end{aligned}$$

By mathematical induction, $P(n)$ is true for all $n > 1$, where $n \in \mathbb{Z}^+$

Question 9

Notes

Since the point M lies on the line, to find k it must satisfy the equation of the line, and also, Parallel line have the same gradient.

(a) Point $M(k, 1)$ lies on line $2x - 14y + 6 = 0$:

$$2k - 14(1) + 6 = 0$$

$$2k - 8 = 0$$

$$k = 4$$

$$\mathbf{k = 4}$$

(b) Length of LM:

$$\begin{aligned} LM &= \sqrt{(4 - (-3))^2 + (1 - 0)^2} \\ &= \sqrt{7^2 + 1^2} \\ &= \sqrt{49 + 1} \\ &= \sqrt{50} \\ &= 5\sqrt{2} \end{aligned}$$

$$\mathbf{5\sqrt{2} \text{ units}}$$

(c) Line $2x + y + 5 = 0$ has gradient -2 .

Parallel line through $M(4, 1)$ with gradient -2 :

$$y - 1 = -2(x - 4)$$

$$y = -2x + 8 + 1$$

$$y = -2x + 9$$

$$\mathbf{y = -2x + 9}$$

Question 10

Notes

Complete the square to find vertex form, which reveals maximum value and vertex coordinates.

(a) Completing the square:

$$\begin{aligned}
 y &= 10x - x^2 - 27 \\
 &= -(x^2 - 10x) - 27 \\
 &= -[(x - 5)^2 - 25] - 27 \\
 &= -(x - 5)^2 + 25 - 27 \\
 &= -(x - 5)^2 - 2 \\
 &= -(5 - x)^2 - 2
 \end{aligned}$$

In the form $-(a - x)^2 + b$:

$$\mathbf{a = 5, \quad b = -2}$$

(b) Since $-(x - 5)^2 \leq 0$ for all x , then:

$$y = -(x - 5)^2 - 2 \leq -2 < 0 \quad \text{for all } x$$

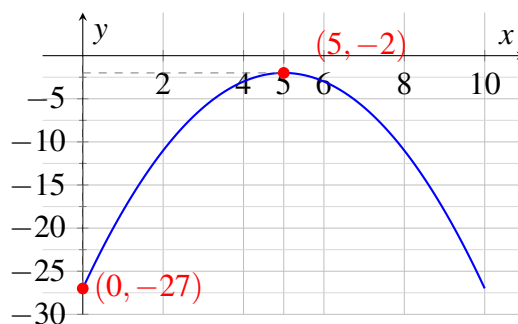
Thus $10x - x^2 - 27$ is always negative.

(c) Maximum point occurs when $(x - 5)^2 = 0$, i.e., $x = 5$:

$$y = -0 - 2 = -2$$

$$\mathbf{(5, -2)}$$

(d) Sketch:



Question 11

Notes

Perform polynomial division to get proper form, then identify transformations from $y = 1/x$.

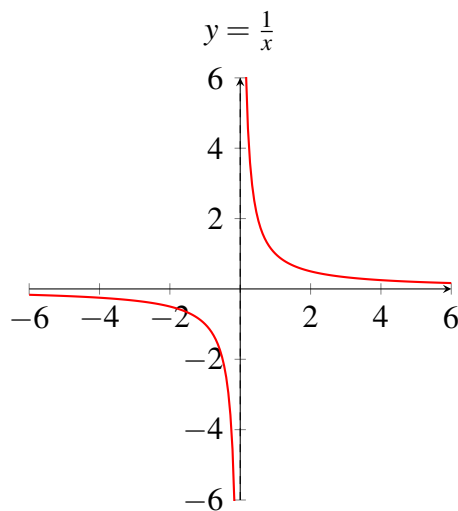
(a) Polynomial division:

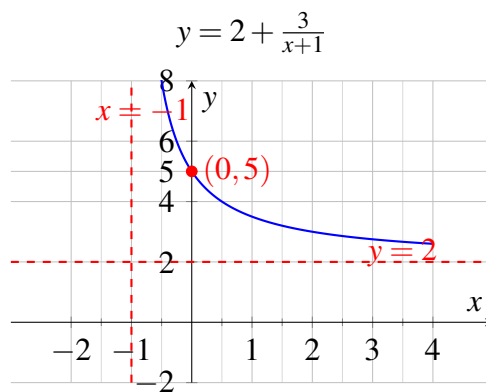
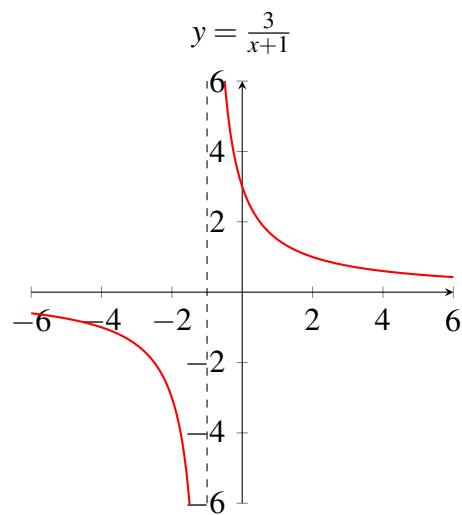
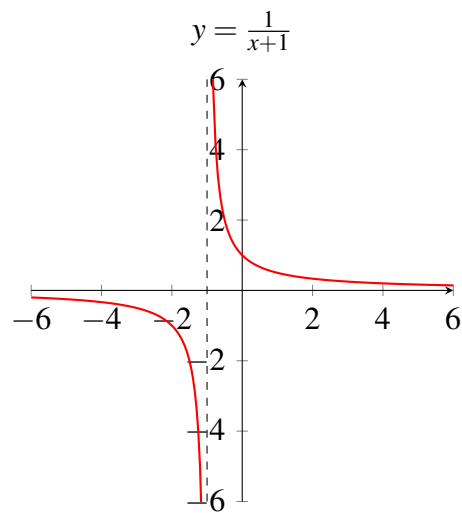
$$\begin{aligned}\frac{2x+5}{x+1} &= \frac{2(x+1)+3}{x+1} \\ &= 2 + \frac{3}{x+1}\end{aligned}$$

(b) Starting from $y = \frac{1}{x}$:

1. Translation by vector $\begin{pmatrix} -1 \\ 0 \end{pmatrix}$: $y = \frac{1}{x+1}$
2. Vertical stretch scale factor 3: $y = \frac{3}{x+1}$
3. Translation by vector $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$: $y = 2 + \frac{3}{x+1}$

(c) Sketch:

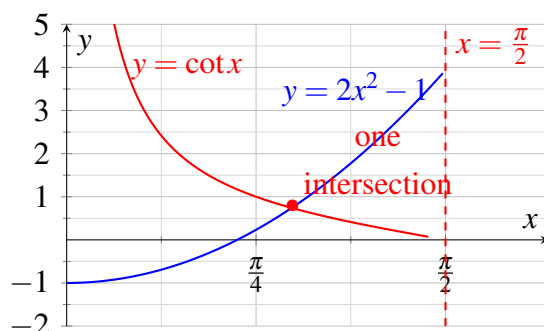




Question 12

Notes

Sketch $y = \cot x$ and $y = 2x^2 - 1$ on the same axes to show exactly one intersection in $(0, \pi/2)$.



Question 13

Notes

Check all four group axioms: closure, associativity, identity, inverses. Create Cayley table for addition modulo 4.

Addition modulo 4 table:

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Closure: All results are in $\{0, 1, 2, 3\}$, hence closed.

Associativity: $(0 + 1) + 2 = 1 + 2 = 3$ and $0 + (1 + 2) = 0 + 3 = 3$, hence Associative.

Identity: 0 is identity since $a + 0 = a = 0 + a$ for all a

Inverses:

- $0^{-1} = 0$ since $0 + 0 = 0$
- $1^{-1} = 3$ since $1 + 3 = 4 \equiv 0 \pmod{4}$
- $2^{-1} = 2$ since $2 + 2 = 4 \equiv 0 \pmod{4}$
- $3^{-1} = 1$ since $3 + 1 = 4 \equiv 0 \pmod{4}$

$(G, +_4)$ is a group under addition modulo 4 as it satisfies all group axioms.

Question 14

Notes

Find inverse by swapping x and y and solving. For composition, substitute $g(x)$ into $f(x)$.

(a) (i) For $f(x) = e^{3x}$:

$$y = e^{3x}$$

$$\ln y = 3x$$

$$x = \frac{1}{3} \ln y$$

$$f^{-1}(x) = \frac{1}{3} \ln x$$

Domain: $x > 0$ (since $\ln x$ requires positive argument)

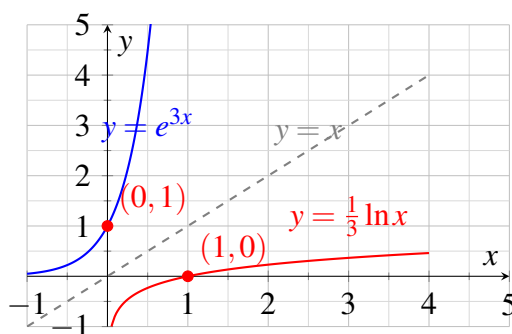
$$f^{-1}(x) = \frac{1}{3} \ln x, \quad x > 0$$

(ii) Composite function:

$$\begin{aligned} fg(x) &= f(g(x)) = f(\ln(x+2)) \\ &= e^{3\ln(x+2)} \\ &= e^{\ln(x+2)^3} \\ &= (x+2)^3 \end{aligned}$$

$$fg(x) = (x+2)^3$$

(b) Sketch for $f(x)$ and $f^{-1}(x)$

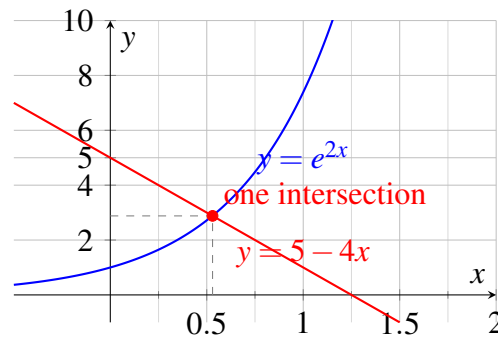


Question 15

Notes

Graphical analysis, sign change verification, then Newton-Raphson iteration.

(a) Graphical approach:



Two graphs intersect at only one point, hence exactly one root.

(b) Sign change method: Let $f(x) = e^{2x} + 4x - 5$

$$f(0) = e^0 + 0 - 5 = 1 - 5 = -4 < 0$$

$$f(1) = e^2 + 4 - 5 = e^2 - 1 \approx 7.389 - 1 = 6.389 > 0$$

Sign change indicates root in $(0, 1)$.

(c) Newton-Raphson method:

$$f(x) = e^{2x} + 4x - 5, \quad f'(x) = 2e^{2x} + 4$$

$$\text{Formula: } x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

With $x_0 = 0.5$:

$$f(0.5) = e^1 + 2 - 5 = 2.71828 + 2 - 5 = -0.28172$$

$$f'(0.5) = 2e^1 + 4 = 5.43656 + 4 = 9.43656$$

$$x_1 = 0.5 - \frac{-0.28172}{9.43656} = 0.5 + 0.02986 = 0.52986$$

Next iteration:

$$f(0.52986) \approx e^{1.05972} + 4(0.52986) - 5 \approx 2.885 + 2.11944 - 5 = 0.00444$$

$$f'(0.52986) \approx 2e^{1.05972} + 4 \approx 5.77 + 4 = 9.77$$

$$x_2 \approx 0.52986 - \frac{0.00444}{9.77} \approx 0.52941$$

Hence:

$$x \approx 0.53 \quad (2 \text{ decimal places})$$

Question 16

Notes

Find derivative, set to zero for stationary points. Use second derivative to determine nature (max/min).

(a)

$$y = 2x^3 + 3x^2 - 12x + 1$$

$$\frac{dy}{dx} = 6x^2 + 6x - 12$$

$$\frac{d^2y}{dx^2} = 12x + 6$$

Stationary points: $\frac{dy}{dx} = 0$

$$6x^2 + 6x - 12 = 0$$

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0$$

$$x = -2 \quad \text{or} \quad x = 1$$

When $x = -2$:

$$y = 2(-8) + 3(4) - 12(-2) + 1 = -16 + 12 + 24 + 1 = 21$$

$$\frac{d^2y}{dx^2} = 12(-2) + 6 = -24 + 6 = -18 < 0 \quad (\text{maximum})$$

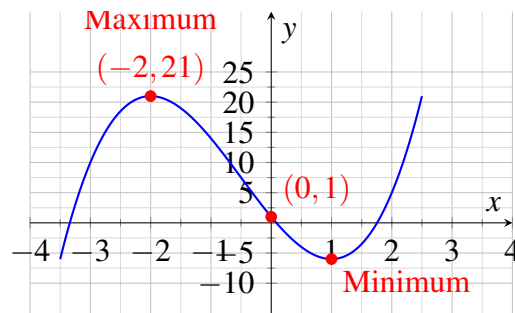
When $x = 1$:

$$y = 2(1) + 3(1) - 12(1) + 1 = 2 + 3 - 12 + 1 = -6$$

$$\frac{d^2y}{dx^2} = 12(1) + 6 = 18 > 0 \quad (\text{minimum})$$

Maximum at (-2, 21), Minimum at (1, -6)

(b) Sketch:



Question 17

Notes

Find determinant, then matrix of cofactors, transpose to get adjoint, divide by determinant.

(a) For $P = \begin{pmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & -2 & 3 \end{pmatrix}$:

Determinant:

$$\begin{aligned} |P| &= 1 \begin{vmatrix} -1 & 1 \\ -2 & 3 \end{vmatrix} - 1 \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} + 2 \begin{vmatrix} 2 & -1 \\ 1 & -2 \end{vmatrix} \\ &= 1[(-1)(3) - (1)(-2)] - 1[(2)(3) - (1)(1)] + 2[(2)(-2) - (-1)(1)] \\ &= 1[-3 + 2] - 1[6 - 1] + 2[-4 + 1] \\ &= 1(-1) - 1(5) + 2(-3) \\ &= -1 - 5 - 6 = -12 \end{aligned}$$

Cofactor matrix:

$$C = \begin{pmatrix} + \begin{vmatrix} -1 & 1 \\ -2 & 3 \end{vmatrix} & - \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} & + \begin{vmatrix} 2 & -1 \\ 1 & -2 \end{vmatrix} \\ - \begin{vmatrix} 1 & 2 \\ -2 & 3 \end{vmatrix} & + \begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix} & - \begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} \\ + \begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix} & - \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} & + \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} \end{pmatrix} = \begin{pmatrix} -1 & -5 & -3 \\ -7 & 1 & 3 \\ 3 & 3 & -3 \end{pmatrix}$$

Adjoint = transpose of cofactor matrix:

$$\text{adj}(P) = \begin{pmatrix} -1 & -7 & 3 \\ -5 & 1 & 3 \\ -3 & 3 & -3 \end{pmatrix}$$

Inverse:

$$P^{-1} = \frac{1}{|P|} \text{adj}(P) = -\frac{1}{12} \begin{pmatrix} -1 & -7 & 3 \\ -5 & 1 & 3 \\ -3 & 3 & -3 \end{pmatrix}$$

(b) Solving $P \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \\ 5 \end{pmatrix}$:

Using P^{-1} :

$$\begin{aligned} \begin{pmatrix} x \\ y \\ z \end{pmatrix} &= P^{-1} \begin{pmatrix} 0 \\ -3 \\ 5 \end{pmatrix} \\ &= -\frac{1}{12} \begin{pmatrix} -1 & -7 & 3 \\ -5 & 1 & 3 \\ -3 & 3 & -3 \end{pmatrix} \begin{pmatrix} 0 \\ -3 \\ 5 \end{pmatrix} \\ &= -\frac{1}{12} \begin{pmatrix} (-1)(0) + (-7)(-3) + (3)(5) \\ (-5)(0) + (1)(-3) + (3)(5) \\ (-3)(0) + (3)(-3) + (-3)(5) \end{pmatrix} \\ &= -\frac{1}{12} \begin{pmatrix} 21 + 15 \\ -3 + 15 \\ -9 - 15 \end{pmatrix} \\ &= -\frac{1}{12} \begin{pmatrix} 36 \\ 12 \\ -24 \end{pmatrix} \\ &= \begin{pmatrix} -3 \\ -1 \\ 2 \end{pmatrix} \end{aligned}$$

$$\mathbf{x = -3, \quad y = -1, \quad z = 2}$$

END OF SOLUTIONS

**FEEL FREE TO CONTACT US FOR ANY ADJUSTMENTS, CLARIFICATIONS AND
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